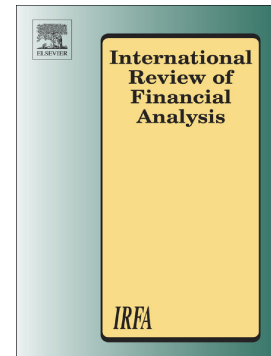


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The effect of green finance on corporate biodiversity risks

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ABSTRACT

This study employs panel data of Chinese listed enterprises from 2000 to 2022 to construct an econometric model examining the mitigating effects of green finance on corporate biodiversity risk and its underlying mechanisms. The empirical results demonstrate that green finance significantly reduces biodiversity risk, with the findings remaining valid across robust. Mechanism analysis reveals that green finance primarily mitigates biodiversity risk by fostering corporate green development, while regional climate risk plays a critical role in shaping the transmission process. Further heterogeneity analysis shows that the risk-mitigating effects are more pronounced in low-pollution industries, firms led by environmentally conscious executives, and regions with stricter environmental regulation. These findings provide novel empirical evidence on the governance role of green finance, offering valuable insights for managing corporate biodiversity risks and refining the green finance framework.

Keywords: Green Finance, Corporate Biodiversity Risk, Corporate Green Development, Regional Climate Risk

1. Introduction

Biodiversity risk, driven largely by human activities such as habitat destruction, overexploitation, pollution, invasive species, and climate change, has emerged as a critical challenge to global sustainability. With the systemic and irreversible nature, it manifests through accelerated species extinction, genetic erosion, and nonlinear ecological tipping points (Rockström et al., 2009, IPBES, 2019). The consequences are profound, undermining ecosystem stability, resilience, and critical services including pollination, water purification, and climate regulation (Cardinale et al., 2012). These disruptions further exacerbate food insecurity, human health crises, and socioeconomic

inequalities, ultimately posing a fundamental threat to global sustainability and human well-being (Steffen et al., 2015). In the view of economics, such degradation exacerbates market failures, threatens supply chain stability, intensifies health and inequality challenges, and ultimately jeopardizes long-term economic development and human welfare.

Corporate survival and growth are deeply reliant on biodiversity and stable ecosystems (Wilting & van Oorschot, 2017). Over 50% of global GDP demonstrates moderate to high functional interdependencies with biodiversity, as quantified by the World Economic Forum reports (WEF, 2020). Yet, biodiversity loss is accelerating worldwide, with one-quarter of all species under threat and approximately one million species facing extinction (NGFS, 2021). Even as human civilization currently faces global urgent risks such as intensified regional conflicts and the proliferation of misinformation and disinformation, concerns regarding extreme weather events, biodiversity loss, and ecosystem collapse have not diminished due to their long-term and gradual nature. Among these, the risk of biodiversity loss has risen to become the third most significant threat in recent World Economic Forum's Global Risk Report.¹

The rapid decline introduces considerable uncertainty and risk to corporate sustainability. Biodiversity loss directly disrupts production and business operations (White et al., 2023), while growing societal pressure from governments, investors and the public demands corporate accountability in biodiversity protection. Entities demonstrating non-compliance with such standards face exposure to reputational capital erosion, compliance-related sanction risks, and competitive disadvantage in capital allocation mechanisms (Smith et al., 2020). Addressing biodiversity risks has, therefore, become essential for corporations. Nevertheless, a substantial funding gap persists in biodiversity conservation and risk management. According to estimates from the Paulson Institute, global biodiversity conservation requires an annual investment of approximately \$711 billion, with current funding levels covering less than 20% of this

¹ See World Economic Forum Global Risks Report 2024 and 2025.

need, predominantly from government sources. This gap underscores two primary implications. Firstly, it calls for increased engagement from both social and private capital, which must shoulder greater responsibilities in this domain (Deutz et al., 2020). Secondly, the efficacy of policy resources in addressing biodiversity risks warrants further empirical investigation. This study constitutes a direct response to the second implication.

Due to the vast territory and rapid development in recent decades, China is one of the world's most biodiverse countries, yet also one where biodiversity faces among the greatest threats. Habitat fragmentation, environmental contamination, and climate stressors, rendering it one of the most biodiversity-endangered countries. To combat that, the central government has institutionalized a multi-tiered governance framework. Firstly, China government Spearheaded the "Kunming-Montreal Global Biodiversity Framework" implementation initiative during the 28th Conference of the Parties (COP28) (CBD, 2022). Secondly, it further revised the National Biodiversity Conservation Strategy and Action Plan (2023-2030) in January 2024, aligning closely with the Kunming-Montreal Framework. These measures persistently advance the integration of biodiversity conservation as a mainstream consideration in national and local policymaking, legal frameworks, and regulatory formulation, while catalyzing all sectors' pragmatic actions to restore and rejuvenate biodiversity (MEPC, 2023). In parallel with integrating biodiversity conservation into regulatory frameworks, the China government has also been promoting its protection through finance innovations. For example, financial institutions are explicitly required to support ecological environment quality improvement and biodiversity conservation. This includes especially considering the value of ecological products to increase support for protection and restoration of critical ecosystems. Furthermore, China is directing public and private capital toward biodiversity-friendly sectors by issuing bonds and establishing funds, and encouraging financial institutions to integrate biodiversity factors into risk management and disclosure frameworks (like LEAP in TNFD). These

measures aim to alleviate funding gaps in biodiversity conservation through financial instrument innovation and policy incentives, thereby advancing the goals of the Kunming-Montreal Global Biodiversity Framework.

More importantly, the green finance framework now incorporates biodiversity protection elements. In 2021, the updated Green Loan Statistical System extended support to biodiversity-focused projects, including ecological restoration and conservation. Banks have developed diversified financial instruments to aid biodiversity protection, such as expanding collateral options, streamlining lending processes and adjusting credit access criteria for biodiversity-related initiatives. Additionally, insurance companies have introduced policies to insure wildlife, further contributing to biodiversity protection. Beyond the financial sector, enterprises across various industries are actively fulfilling environmental responsibilities and implementing ESG principles (Zhang et al., 2025). They pursue multidimensional optimization of decarbonization imperatives and biospheric integrity preservation through synchronous advancement of circular value creation paradigms, culminating in anthroposphere-nature symbiosis (Wang et al., 2024). However, scholarly communities remain deeply divided on whether China's green finance framework has effectively mitigated biodiversity risks, safeguarded species diversity and preserved ecosystem integrity. Therefore, a rigorous investigation is warranted to systematically examine the potential effects and the underlying mechanisms

Based on Chinese listed companies (2000~2022), this study uses Python-based text crawling to extract biodiversity-related keywords and phrases from annual reports. Sentiment analysis is employed to evaluate corporate biodiversity risk exposure, with municipal green finance development quantified through a composite index integrating green credit, securities, and insurance market indicators. A deductive analytical framework is then constructed, and the empirical results confirm that green finance policies exert significant mitigating effects on corporate biodiversity risks. These findings remain robust across a battery of robustness checks. Furthermore, this study

leverages the difference-in-differences quasi-experimental framework embedded in China's Green Finance Reform and Innovation Pilot Zones policy, implementing rigorous exogenous policy shock identification. It empirically validates the biodiversity risk attenuation capacity inherent in green financial mechanisms, establishing causal inference through institutional isomorphic pressures derived from regulatory experimentation. Building on this, the study also identifies and validates the key mediating mechanism by which green finance influences biodiversity risk: fostering corporate green. This study also conducts heterogeneity analyses, and the findings reveal that the risk-mitigating effects of green finance are more pronounced in enterprises operating within industries with lower pollution intensity, and among firms whose executives demonstrate heightened green awareness. Notably, our analysis reveals that climate risk exerts a significant positive moderating effect on the relationship between green finance and corporate green development. This finding offers valuable insights for strengthening the precision and policy relevance of green finance initiatives.

This paper makes several contributions. First, it innovatively integrates theories and methods from finance, ecology, and business management. This integration establishes a unified analytical framework for corporate biodiversity issues that addresses limitations of single-discipline research. The implementation of text analytics tools effectively addresses the quantification challenges inherent in corporate biodiversity risk assessment. Therefore, our works not only provide novel pathways for firms to achieve dual alignment of ecological and economic objectives, but also offer valuable insights to inform the development of scientifically robust and economically viable biodiversity governance strategies.

Second, while previous research on green finance and biodiversity has primarily been theoretical (Seidl et al., 2024; Karolyi & Tobin-de la Puente, 2023), this research advances the knowledge in the field by employing text mining techniques to extract data for regression analysis, seeking to establish robust quantitative evidence for the

emerging nexus between green finance and corporate biodiversity risk exposure. Specifically, leveraging BERT-based text mining of annual reports, we first quantify corporate biodiversity risk exposure, then assess its magnitude through systematic analysis of negative disclosure frequency.

Third, through an analysis of China's listed companies, this study extends the applicability boundaries of green finance while establishing empirical validation for the institutional efficacy of green financial policies in ecological governance. (Huang et al., 2022; Wang et al., 2024).

Finally, by interrogating the mediating pathways and heterogeneous treatment effects of green finance, this investigation elucidates the mechanistic topology and regulatory thresholds of such financial interventions. Such analytical progression enables systematic identification of green finance's marginal regulatory gains and institutional inertia, thereby furnishing both theoretical scaffolding and empirical diagnostics for optimizing the adaptive capacity of green financial architecture.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature, establishes the academic foundation of the study, and then develops the hypotheses accordingly. Section 3 presents the research design, including data sources, variable construction, and model specification, with particular emphasis on the derivation of the core indicators. Section 4 provides the empirical analysis, covering descriptive statistics, baseline regressions, robustness checks, and mediation mechanism tests. Section 5 further extends the analysis by examining moderating effects, supplemented by consistency checks. Finally, Section 6 concludes the paper with a summary of the findings, a discussion of their broader implications, and corresponding policy recommendations.

2. Literature review and research hypothesis

2.1. Literature Review

Two strands of literature are particularly illuminating for this study. The first concerns the evolution of green finance, tracing its development from addressing

climate change to encompassing biodiversity protection. The second focus on the impact of biodiversity risk on enterprises.

2.1.1 The evolution and effects of green finance

Green finance initially emerged to tackle climate change, focusing on mitigation and adaptation. Its core objectives were managing climate-related financial risks and directing capital toward low-carbon sectors. Banga (2019) conceptually linked the growth of green finance to climate risk hedging, finding that investors increasingly perceive green assets as a hedge against transition risks. Subsequently, Flammer (2021) provided robust empirical evidence demonstrating that the issuance of green bonds leads to a significant reduction in issuers' carbon emissions profile, validating the channel of capital reallocation. Earlier, Tang & Zhang (2020) had proved that green credit policies effectively curb carbon intensity in heavy-polluting borrowers by raising their financing costs and incentivizing technological upgrades. After that, a growing body of evidence underscores the intrinsic link between the climate change and biodiversity crises, they are rise together, and fall together (IPBES-IPCC, 2021; Seddon et al., 2021). Consequently, the conceptual scope of green finance is expanding beyond its singular focus on enabling a low-carbon transition toward a broader mandate of fostering harmonious coexistence between humanity and natural ecosystems. This shift has been galvanized by international agreements and consensus, most notably the Kunming-Montreal Global Biodiversity Framework (GBF) (2022). The GBF explicitly calls for mobilizing all resources, including financial resources. Then, a comprehensive guide for financial institutions, outlining practical approaches to assess their impact on biodiversity and align their portfolios with the goals of the GBF (FBI, 2023). China has also integrated biodiversity into its green finance policy framework, including updating its Green Bond Endorsed Projects Catalogue and carrying out Green Finance Reform and Innovation Pilot Zones, and so on. Zhang et al. (2025) found that the establishment of Green Finance Reform and Innovation Pilot Zones in China led to a measurable increase in local bird species richness, demonstrating a positive impact on biodiversity.

Referring to biodiversity-themed bond financing in China, CCDC (2025) provided evidence of market uptake and policy effectiveness.

2.1.2 The corporate biodiversity risk and economic impacts

The biodiversity risk is widely recognized to exert substantial impacts on macroeconomic development and policy formulation, including straining national fiscal balances, threatening global supply chain stability, and undermining sustainable development (IPBES, 2019). The impact of biodiversity risk on stock price crashes and its consequent threat to financial market stability has emerged as a prominent research focus in recent years (Liang et al., 2024; Bassen et al., 2024). Due to the pioneering work by Giglio et al. (2023) in the development of quantitative metrics, an increasing number of scholars are focusing on biodiversity risk issues at the micro-enterprise level (Li et al., 2025; Hasan et al., 2025). Corporate biodiversity risk refers to financial risks arising from biodiversity loss and ecosystem degradation (Giglio et al., 2023). Building on the climate risk taxonomy, scholars classify it into physical risks (e.g., operational disruptions caused by degraded ecosystem services) and transition risks (e.g., rising compliance costs due to policy changes). Biodiversity risk influences enterprise's operational and financial stability through three primary channels. The first one is stringencing regulatory compliance cost. Stricter environmental regulations compel firms to increase expenditures on pollution control, ecological restoration, and environmental monitoring, directly raising operational costs (Adler et al. 2017). The second one is rising financing cost. Financial institutions are integrating biodiversity risks into credit pricing, elevating borrowing costs for exposed firms through higher interest rates and stricter terms (Li et al., 2025). The third one is disrupting chain supply stability. Ecosystem degradation disrupts raw material supply stability for agriculture, pharmaceuticals, and energy sectors, causing operational delays and procurement risks (Adamolekun, 2024). For corporations, biodiversity risk most directly manifests in their financing costs, with mounting evidence indicating that this risk is being accurately priced in by equity markets, bond markets, and the banking system (Cherief et al., 2025;

Erten & Ongena, 2024; Li et al., 2025; Liang et al., 2024).

In summary, while biodiversity risk regulations are intensifying, it must be concurrently acknowledged that supporting policies have been progressively refined. Consequently, whether and how current green finance policies influence corporate biodiversity risk constitutes a topic of both theoretical and practical significance. Given the underexplored nature of this issue in extant literature, this study endeavors to develop robust and well-grounded logical relationships between these mechanisms.

2.2 Research Hypothesis

2.2.1 Green Finance and Corporate Biodiversity Risks

Green finance provides financial services, including green credit, bonds, insurance and funds, that help environmental improvement, climate action, and resource efficiency initiatives (Bahl, 2012). Green finance affects corporate biodiversity risk mainly through mechanisms of incentivization, constraint and oversight, guiding firms toward sustainable practices that reduce biodiversity threats.

Firstly, green finance stimulates companies to invest in biodiversity conservation and environmental protection projects through various preferential policies and financial instruments. These policies effectively lower corporate capital costs while generating sustainable economic growth and ecological conservation benefits over extended time horizons (Huang et al., 2021), motivating corporate participation in conservation and thus reducing biodiversity risks. At present, some banking institutions are aligning green finance products with biodiversity initiatives, introducing targeted green loans and bonds that support nature reserves, wetland restoration and similar projects. Leading insurance providers have expanded their offerings, covering costs such as reforestation for damaged forests and providing liability insurance for fisheries affected by accidents and pollution. Some institutions have innovated programs, which incorporate biodiversity-related metrics into sustainability-linked loans and bonds.

Secondly, green finance imposes restrictions on high-pollution, high-energy-

consuming companies and those failing to meet environmental standards, utilizing mechanisms within the financial market (Scholtens, 2017). Many financial institutions have incorporated biodiversity indicators into corporate evaluation criteria. For non-compliant companies, financial institutions may impose higher financing costs, limit loan access or even refuse funding, pressuring companies to address biodiversity risks through active risk management.

Lastly, green finance mandates have enhanced corporate transparency, requiring regular disclosure of biodiversity initiatives and environmental risk assessments (Komendantova, 2021). This fosters stronger oversight from the public, investors and regulators, prompting companies to improve their environmental management and accountability. Through transparent reporting and public scrutiny, companies have increasingly prioritized sustainable development, reducing potential biodiversity threats. Building on the systematic examination of preceding evidence, we advance the following research hypothesis:

Hypothesis 1. Green finance mechanisms can alleviate enterprise-level biodiversity risk exposure.

2.2.2 The Mediating Role of Corporate Green Practices

Green financial policies, by integrating environmental protection as a core principle, establish and enforce comprehensive sustainability standards encompassing ecological conservation, resource efficiency, and climate resilience. These measures provide structured pathways for corporate green transition and sustainable development. These standards encompass various areas, including environmental impact assessments, green technology resource and development, and resource recycling, which support companies in identifying green risks, optimizing operations, and seizing green development opportunities (Xing et al., 2021). In the meantime, corporate green development critically relies on innovation in green technologies (Hu et al., 2021). Given the high capital intensity and elevated risk exposure with such innovations, green finance supports this process by offering diverse financial tools that lower capital costs

and mitigate risks through shared-risk mechanisms (Lee & Lee, 2022). This dual impact significantly enhances corporate engagement in green innovation, fostering green technology output and expediting green development.

In parallel, corporate green development directly mitigates biodiversity risks. By adopting more efficient, environmentally friendly technologies, green-focused companies reduce resource over-exploitation and environmental degradation, safeguarding biological habitats and reducing dependence on external ecosystems (Zhao et al., 2024). Furthermore, strategic green practices, such as green procurement, cleaner production, sustainable packaging and low-carbon logistics, spread through the supply chain, influencing suppliers, manufacturers, distributors and consumers. This diffusion mechanism not only promotes green technology exchange but also encourages proactive green practices across the supply chain (Li et al., 2017). Consequently, corporate green development sets a model for sustainable practices, reducing the overall ecological footprint of the supply chain and, thereby, lowering biodiversity risks. Additionally, with the rise in environmental awareness, corporate green development increasingly aligns with regulatory standards and meets the expectations of investors and consumers (Plumlee et al., 2015), effectively reducing the associated transitional risks. Derived from those, the study formally posits a research hypothesis as:

Hypothesis 2. Green finance mitigates corporate biodiversity risk through its role in advancing corporate green development.

3. Research Design

3.1. Data

3.1.1. *Corporate biodiversity risk*

The research sample encompasses China's A-share listed enterprises spanning 2000-2022, with firm-specific characteristics and financial data strategically acquired from the China Stock Market & Accounting Research (CSMAR) Database. Biodiversity risk

metrics were systematically extracted via computational content analysis of corporate annual reports.

3.1.2. Green finance development

The original green finance index quantitatively derived from EPS Global and WIND Economic Database. Specifically, conforming to established research conventions, we first collected raw data pertaining to green credit, green investment, green insurance, green bonds, green support, green funds and green equity across prefecture-level administrative units. Then the Entropy Weight Method (EWM) is applied to derive objective coefficient weights across all dimensions. The details is in Table 1.

3.1.3. Others

Climate risk indicators compiled from the annual China Climate Change Monitoring Bulletin. The data of corporate green tax incentives is as well from CSMAR. Following rigorous data integration, three-stage filtering was implemented: (1) exclusion of firms with ST or *ST status in the observation year; (2) removal of financial institutions and real estate companies; (3) comprehensive 1% winsorization at both extremes for continuous variables. These procedures yielded a final sample of 3,664 firm-year observations.

3.2. Variable Definitions

3.2.1. Dependent Variables

Corporate Biodiversity Risk (*BR*). Building upon prior work of Liang et al. (2024) and Ma et al. (2024), this study quantifies corporate biodiversity risk (*BR*) by analyzing the sentiment of biodiversity-related contexts in corporate annual reports and integrating both physical risks and transition risks. This indicator incorporates both physical and transition risks associated with biodiversity. Physical risks refer to the direct impact on corporate operations due to biodiversity loss, while transition risks involve potential corporate closures, defaults and other threats arising from government policies aimed at biodiversity protection.

To precisely capture these risk signals, the study constructed a biodiversity lexicon and a regulatory lexicon to identify sentences related to biodiversity's physical and transition risks in annual reports. Following the identification of contextually relevant statements and removal of extraneous content, all biodiversity-related sentences were subjected to sentiment analysis using the BERT model, with classification into positive, neutral, and negative sentiment categories. The company's expressed attitude toward biodiversity topics serves as an indicator of its potential risk exposure. The net count of negative versus positive sentiment sentences serves as the core measure of corporate biodiversity risk. This measure provides a direct reflection of the level of negative attention to biodiversity issues in annual reports. A higher value indicates greater biodiversity risk for the company.

3.2.2. Independent Variables

As instruments can reflect policy effectiveness (Zhang et al., 2023), the Green Finance Development Index (*GFI*) in this study is constructed through a multidimensional evaluation framework. Expanding the analytical framework established by Lv et al. (2021), this study operationalizes seven granular metrics to evaluate the *GFI* within corporate operational jurisdictions, with jurisdictional boundaries defined at the prefectural administrative level. Table 1 delineates the methodological specifications of each metric's operationalization. A multivariate statistical procedure employing information entropy principles. Specifically, the Entropy Weight Method (EWM) is algorithmically applied to derive objective coefficient weights across all dimensions.

Table 1 Green Finance Development Index System

Sub-indicator	Measurement
Green Credit	The ratio of total credit for environmental projects and regional credit aggregate
Green Investment	The ratio of environmental pollution control investment and regional GDP
Green Insurance	The ratio of environmental pollution liability insurance premiums and total premium income

Green Bonds	The ratio of green bond issuance volume and total bond issuance volume
Green Support	The ratio of fiscal expenditure on environmental protection and local general budgetary expenditure
Green Funds	The ratio of market value of green funds and market value of all funds
Green Equity	(Carbon trading + energy rights trading + emission rights trading) / Total equity market trading volume

3.2.3. Mediation Variables

Green Development (*GD*), as a significant trend in today's economic landscape, is evidenced by enhanced green total factor productivity (GTFP). Extending the work of Wu et al. (2020), the GTFP, measured by the non-radial Slack-Based Measure Malmquist-Luenberger (SBM-ML), is introduced as the key indicator for evaluating corporate development levels.

3.2.4. Control Variables

This study employs a multivariate control framework anchored in established corporate governance and financial econometrics literature. Extending the methodological architecture proposed by Lv et al. (2021), we operationalize eight multidimensional control parameters, including Firm Size (*Size*), Financial Leverage (*Lev*), Return on Equity (*ROE*), Liquidity Ratio (*Quick*), Board Size (*Board*), Loss-Making Status (*Loss*), CEO duality (*Dual*) and managerial Ownership (*Mshare*). additionally, we will also control the time-fixed and industry-fixed effects. Descriptions of each variable are presented in Table 2.

Table 2 Variable Description.

Variables	Descriptions
Dependent Variable	
<i>BR</i>	calculated as the difference between Negative Impact Disclosures and Positive Engagement Statements in corporate filings
Independent Variable	
<i>GFI</i>	Comprehensive index of seven sub-indicators measured by EWM
Mediating Variables	
<i>GD</i>	Green total factor productivity
<i>CR</i>	Constructed by normalizing four sub-indicators

Control Variables

<i>Size</i>	Log-transformed Year-End Total Assets (Ln)
<i>Lev</i>	computed as Total Liabilities scaled by Total Assets at fiscal year-end
<i>ROE</i>	computed as Net Income scaled by Average Shareholders' Equity during the fiscal year
<i>Quick</i>	computed as Liquid Current Assets scaled by Current Liabilities at fiscal year-end, where liquid assets exclude inventories
<i>Board</i>	Log-transformed Year-End board members (Ln)
<i>Loss</i>	1 indicates negative net profit exists, otherwise 0
<i>Dual</i>	1 indicates concurrent chairman-CEO position tenure at fiscal year-end, otherwise 0
<i>Mshare</i>	Equity Holdings of Key Governance Actors (directors, supervisors, and executives) / Total Outstanding Shares

3.3. Research models

To examine how green finance affects corporate biodiversity risk exposures, the econometric framework is structured as follows:

$$BR_{i,t} = \alpha_0 + \alpha_1 GFI_{i,t} + \gamma Controls_{i,t} + Year_t + Industry_i + \varepsilon_{i,t} \quad (1)$$

In Equation (1), the subscripts i and t denote firm and year respectively. $BR_{i,t}$ represent corporate biodiversity risk, $GFI_{i,t}$ indicate the host city's green finance development, $Controls_{i,t}$ symbolized a vector of control variables, $Year_t$ and $Industry_i$ control for time-fixed and industry-fixed effects, with $\varepsilon_{i,t}$ denoting the idiosyncratic error terms.

To further examine the mediation effects of corporate green development and climate risk, a mediation model deriving from Jiang (2021) is constructed as:

$$GD_{i,t} = \alpha_0 + \alpha_1 GFI_{i,t} + \gamma Controls_{i,t} + Year_t + Industry_i + \varepsilon_{i,t} \quad (2)$$

Equation (2), $GD_{i,t}$ denotes corporate green development, other components are as defined in Equation (1).

To further examine the robustness of Hypothesis 1, this study, following Lu et al. (2022), uses the quasi-natural experiment of green finance policy pilots to construct the following DID model.

$$BR_{i,t} = \alpha_0 + \alpha_1 did_{i,t} + \gamma Controls_{i,t} + Year_t + Industry_i + \varepsilon_{i,t}$$

(3)

In Equation (3), the dummy variable $did_{i,t}$ is operationalized as a dichotomous policy variable assuming unity for city i from year t onward upon inclusion in the Green Finance Innovation Pilot Zone roster, and 0 for pre-designation periods.

4. Empirical Results

4.1. Data Description

Prior to measuring corporate biodiversity risk, we draw on the methodology of He et al. (2024) to initiate our analysis by identifying firms' attention to biodiversity issues. Specifically, a firm is assigned a value of 1 if its annual report discloses biodiversity-related content, and 0 if no relevant keywords from our predefined dictionary are identified. Figure 1 illustrates the trend from 2021 to 2023. The left vertical axis shows the proportion of listed firms in China demonstrating biodiversity awareness relative to the total number of listed companies, represented by grey bars. The data reveal a substantial increase in biodiversity awareness among Chinese listed companies, with the proportion rising sharply from 7.63% in 2021 to 74.10% in 2023. The right vertical axis indicates the number of firms: the solid orange line represents the total number, while the dashed blue line depicts the number of firms with biodiversity awareness. By 2023, the total number of listed companies exceeded 5,000, among which 3,869 firms explicitly addressed biodiversity in their annual reports. This indicates that over 70% of listed firms have recognized the intrinsic relationship between biodiversity issues and corporate development.

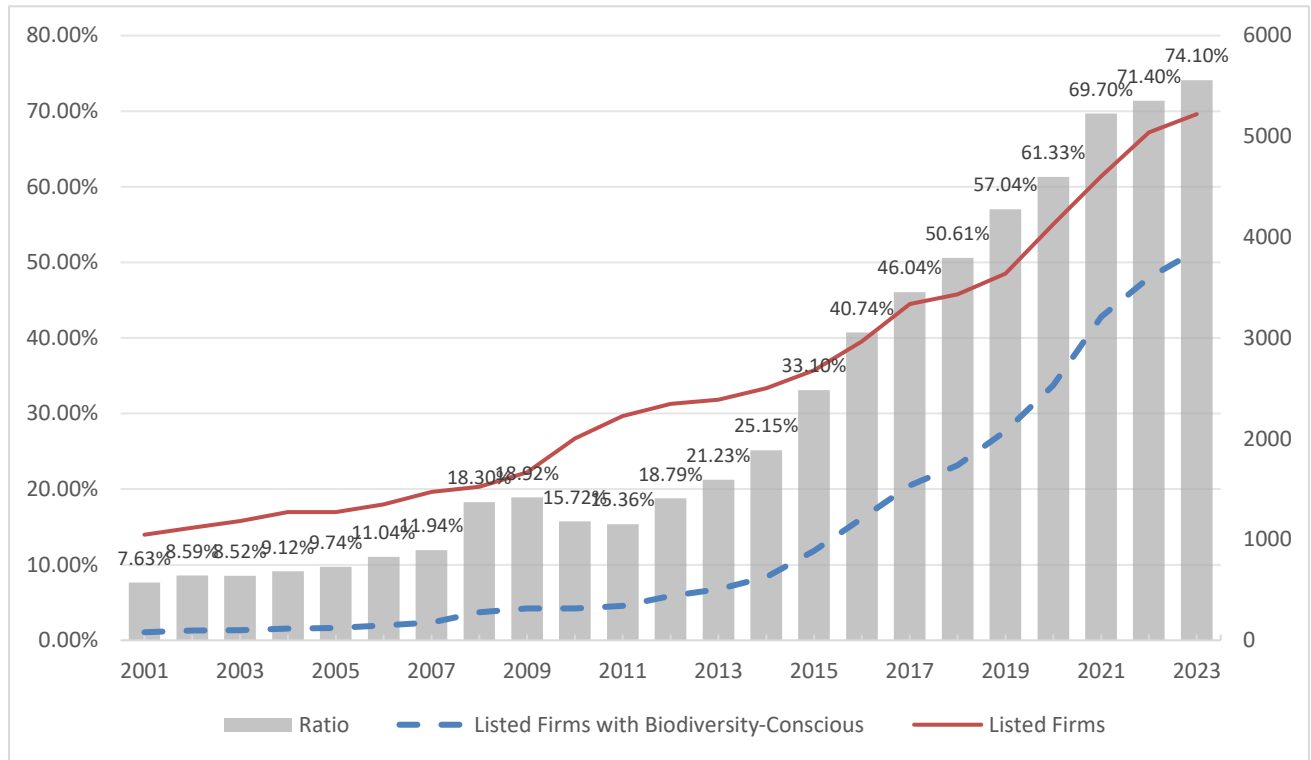


Figure 1. The trend from 2021 to 2023

Furthermore, within the overarching upward trend in biodiversity awareness, three distinct anomalous surges occurred during the observed period. The first surge took place in 2008, when the number of firms explicitly addressing biodiversity increased sharply from 176 to 279, raising the proportion from 11.94% to 18.30%. This shift is potentially attributable to the 2008 financial crisis, which exposed critical corporate governance failures, including short-termism, excessive risk-taking, misaligned incentives, and risk management deficiencies, prompting global reflections and reforms in corporate practices. This period marked the awakening of ESG investment consciousness in the post-crisis era (Kirkpatrick, 2009; Lins et al., 2017). The second surge occurred in 2015, with the share of biodiversity-aware listed companies surpassing 33.10%. This development was likely driven by the positive influence of the United Nations Sustainable Development Goals (SDGs) released that year on Chinese firms (Zhang et al., 2020). The third surge happened in 2021, with nearly 70% of listed companies incorporating biodiversity issues in their annual reports. This suggests that the global COVID-19 pandemic significantly heightened Chinese firms' awareness of

the biodiversity dimensions underlying highly infectious diseases (Broadstock, 2021).

Nevertheless, it must be acknowledged that corporate attention to biodiversity in disclosures does not necessarily reflect genuine concern for biodiversity risks. This is empirically observable in the discrepancy between the initial sample and the final sample after applying frequency-based filtering. Specifically, following the approach of Liang et al. (2024), we only retained reports in which biodiversity-related keywords appeared in at least two distinct sentences. This criterion suggests that a considerable number of Chinese firms may engage in tokenistic disclosure, indicating a lack of substantive understanding of the issue. Further supporting this interpretation, we calculated the mean ratio of keyword characters to the total characters in the report based on the method of He et al. (2024), which yielded a mean value of 0.000156. Given that the average length of annual reports among Chinese listed firms is approximately 40,000 characters (Wang & Chen, 2021), this implies an average of only about six characters devoted to biodiversity topics. Such minimal textual engagement is quantitatively insufficient for meaningful sentiment or content analysis. However, this limitation also underscores a direction for future research.

4.2. Descriptive Statistics

Descriptive statistics summary is reported in table 3. Biodiversity Risk (*BR*) exhibiting mean value of -1.455, suggesting that companies generally demonstrate a predominance of optimism over concern regarding the biodiversity risks they face. However, extreme values (Max=7, Min=-55) indicate that some enterprises face significant biodiversity challenges. The Green Finance Index (*GFI*) demonstrates a range of 0.085 to 0.633 (SD=0.125), revealing significant geographical disparities in municipal green finance policy frameworks. The statistics of the control variables are broadly aligned with those reported in existing literature. For example, the mean of *Loss* is 0.105, which means 10.5% of listed firms have experienced financial losses. *Dual* is 0.224, indicating 22.4% of firms are ownership-control convergence. In terms

of *Mshare*, although the average level is modest (Mean=10.452), there exists significant variation (SD=17.530) across firms.

Table 3 Descriptive Statistics

Variables	N	Mean	SD	Min	p50	Max
<i>BR</i>	3664	-1.455	3.432	-55.000	-1.000	7.000
<i>GFI</i>	3664	0.405	0.125	0.085	0.423	0.633
<i>Size</i>	3664	22.991	1.713	18.287	22.709	28.636
<i>Lev</i>	3664	0.478	0.200	0.020	0.484	1.345
<i>ROE</i>	3664	0.074	0.195	-3.836	0.083	3.499
<i>Quick</i>	3664	1.556	2.209	0.027	1.030	47.37
<i>Board</i>	3664	2.148	0.224	1.386	2.197	2.890
<i>Loss</i>	3664	0.105	0.307	0	0	1.000
<i>Dual</i>	3664	0.224	0.417	0	0	1.000
<i>Mshare</i>	3664	10.452	17.530	0	0.129	75.326

4.3. Baseline Regression Analysis

Considering China's demonstrated historical efficacy in energy conservation and emission reduction policies (Zhang et al., 2024), can similar policy effectiveness be achieved in mitigating biodiversity risks? Table 4 presents the baseline regression estimates assessing the impact of green finance policy implementation on corporate biodiversity risk exposure. This paper focuses on establishing causal identification from the green finance development and to corporate biodiversity risk. We therefore employ a stepwise regression approach for rigorous verification, with a battery of robustness checks to follow. Table 4 demonstrates the baseline regression results.

The unadjusted model (Column 1) demonstrates a statistically significant negative association between the Green Finance Index (*GFI*) and biodiversity risk (*BR*) at the 1% level. This relationship persists in Column 2 after introducing time-fixed and industry-fixed effects, with the *GFI* coefficient remaining robustly negative (10% significance). The fully specified model in Column 3, incorporating both fixed effects and control variables, further confirms *GFI*'s significant negative influence (10% level), establishing result robustness across progressive specifications. These consistent findings validate Hypothesis 1, suggesting that green finance mitigates biodiversity

risks.

Regarding control variables, we find that larger board size significantly reduces corporate biodiversity risk (-0.734***). Conversely, higher managerial ownership intensifies such risk (0.007**). Expanded board composition incorporates broader stakeholder perspectives. This facilitates timely information transmission to senior executives. Consequently, it improves corporate awareness and decision-making on biodiversity issues. However, elevated managerial equity alignment creates countervailing effects. It may strengthen executives' short-term private incentive motives. Agents consequently attenuate the perception of biodiversity exposures. This induces systematic underestimation of ecological risks in strategic choices.

Table 4 Baseline Regression and Mediation Analysis Results

Variables	(1) <i>BR</i>	(2) <i>BR</i>	(3) <i>BR</i>	(4) <i>GD</i>	(5) <i>CR</i>
<i>GFI</i>	-3.397*** (-7.927)	-0.879* (-1.805)	-0.896* (-1.783)	0.052* (1.859)	-11.299** (-2.532)
<i>Size</i>			-0.438*** (-9.998)	-0.002 (-1.067)	0.313 (1.203)
<i>Lev</i>			0.209 (0.520)	0.020** (2.179)	-2.320 (-1.635)
<i>ROE</i>			-0.170 (-1.271)	0.016** (2.028)	-0.112 (-0.482)
<i>Quick</i>			-0.034 (-1.151)	0.001 (0.810)	0.169 (1.632)
<i>Board</i>			-0.734*** (-2.712)	-0.002 (-0.278)	-0.603 (-0.598)
<i>Loss</i>			-0.012 (-0.062)	0.007** (2.006)	0.458 (1.075)
<i>Dual</i>			0.014 (0.101)	-0.001 (-0.384)	-0.133 (-0.281)
<i>Mshare</i>			0.007** (1.961)	0.000 (0.284)	-0.014 (-0.700)
IND FE	NO	YES	YES	YES	YES
Year FE	NO	YES	YES	YES	YES
Observations	3,664	3,664	3,664	3,664	3,664
Adjust R ²	0.0479	0.0869	0.126	0.0164	0.436

Note: ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

The same notation applies to the remaining tables.

4.4. Robustness Checks

To reinforce the validity of our core findings, this section systematically addresses potential empirical concerns through five robustness checks. Each test targets distinct threats to identification: 4.4.1 aims to mitigate confounding effects from exogenous policy shocks (the Kunming Declaration's regulatory impact); 4.4.2 controls for unobserved spatial heterogeneity in regional economic and ecological conditions; 4.4.3 verifies result consistency under a binary response variable framework; 4.4.4 introduces Propensity Score Matching (PSM) to reduce selection bias arising; and 4.4.5 leverages quasi-experimental variation from the Green Finance Pilot Zones to strengthen causal inference. Collectively, these tests corroborate the robustness of *GFI*'s biodiversity risk mitigation effect across model specifications and identification strategies.

4.4.1. Sample Adjustment

The Kunming Declaration, a key outcome of COP15, is an influential document in biodiversity conservation, and its release may prompt policy changes, such as stricter environmental regulations and higher biodiversity protection standards. These policy changes may directly influence corporate biodiversity risk exposure level, thereby introducing interference effects that compromise the operational efficacy of green finance mechanisms. To enhance the robustness of the study's findings, we here exclude post-Kunming Declaration observations and re-conducts regression test. As shown in Column (1) of Table 5, the coefficient of *GFI* remains significantly negative at the 5% level, indicating that the core conclusion holds even after controlling for exogenous policy shock.

4.4.2. Additional Fixed Effects

Significant regional disparities in economic development, policy environment, and natural resource endowments may critically shape corporate biodiversity risk exposure.

To mitigate omitted variable bias and enhance model accuracy, this study incorporates region-fixed effects alongside pre-existing controls for time-fixed and industry-fixed effects. As demonstrated in Column (2) of Table 5, the coefficient of *GFI* remains statistically significant at the 1% level with a negative sign, confirming the robustness of core finding after accounting for spatial heterogeneity.

4.4.3. Alternative Regression Specification

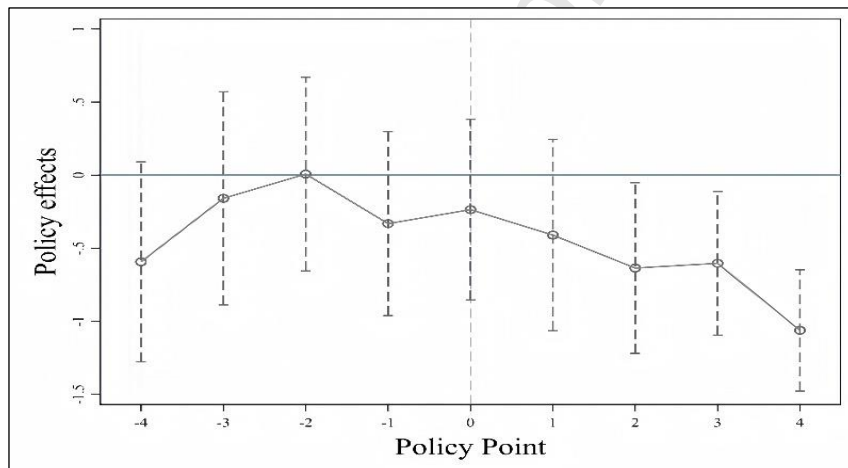
To establish empirical robustness given the categorical distribution of the response variable, we implement a dichotomous transformation *Br* operationalizing biodiversity risk threshold: *Br*=1 for entities with measurable risk exposure, *Br*=0 otherwise. Employing a logit regression model for *Br*, Column (3) of Table 5 demonstrates that *GFI* coefficient retains significance at the 10% level with a negative estimate, substantiating the persistence of core findings following model specification.

Table 5 Results of Robustness Tests

Variables	(1) <i>BR</i>	(2) <i>BR</i>	(3) <i>Br</i>	(4) <i>BR</i>	(5) <i>BR</i>
<i>GFI</i>	-1.313** (-2.566)	-7.421*** (-5.013)	-0.932* (-1.666)	-1.149** (-1.999)	
<i>did</i>					-0.637*** (-4.313)
<i>Controls</i>	YES	YES	YES	YES	YES
IND FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Area FE	NO	YES	NO	NO	NO
Observations	2,422	3,657	3,664	2,832	3,657
Adjust R ²	0.170	0.173	0.173	0.160	0.153

4.4.4. Propensity Score Matching (PSM)

Endogeneity issues arising from omitted variables, reverse causality, and other unobserved confounders may bias estimation results. To mitigate these biases and enhance causal identification, we implement propensity score matching (PSM) analysis. The procedure unfolds as follows: Firms are classified into treatment ($GFI \geq \text{median}$) and control groups ($GFI < \text{median}$). Then, a 1:1 nearest-neighbour matching is performed using a Probit model, with the same covariates as in regression Equation (1). As shown in Figure 2, the differences among covariates are significantly reduced post-matching, indicating effective matching. Finally, the matched sample is re-estimated. The significantly negative coefficient of GFI at the 5% level in Column (4) of Table 5



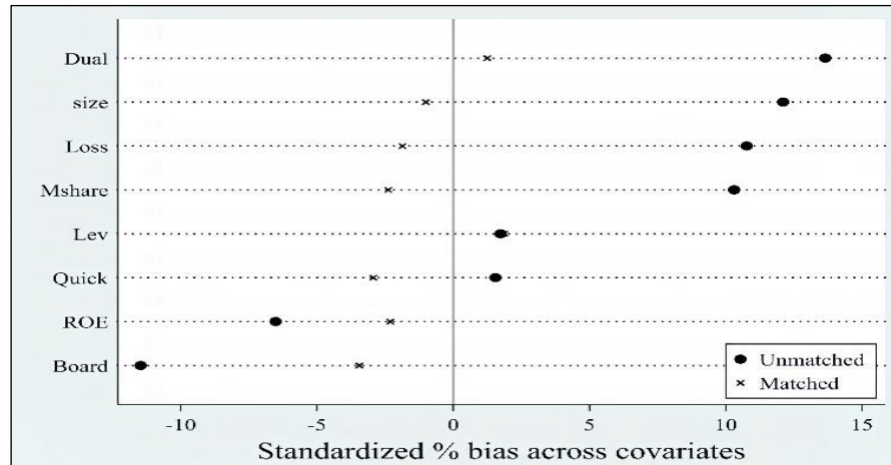
demonstrates that our core findings remain robust even after rigorously addressing endogeneity through advanced model controls.

Figure 2. PSM Balance Plot

4.4.5. Exogenous Shock Analysis

The Green Finance Reform and Innovation Pilot Zone policy was designed to explore novel pathways and models for advancing green finance, enabling financial resources to better support green economic development and achieve both environmental and economic benefits. Zhang et al (2023) has found that this policy promotes energy savings. Therefore, it may also affect the corporate biodiversity risks. Here, this policy serves as an ideal quasi-natural experiment for our analysis. Accordingly, following

with the endogeneity treatment methodology of Fuchs-Schündeln & Hassan (2016), we treat the Green Finance Reform and Innovation Pilot Zone² as an exogenous policy shock, employing a DID framework to conduct the evaluation. Prior to regression analysis, we conduct parallel trend and dynamic effect test using standard event-study methodology, which confirms the suitability of the DID estimation (see Figure 3).



Following that, we proceed with regression analysis using Equation (3). The Column (5) of Table 5 shows a significantly negative *DID* coefficient, further supporting the conclusion that green finance can mitigate a firm's biodiversity risk.

Figure 3. Parallel Trends Test Plot

4.5. Mediation Analysis

The classical approach to testing mediating effects typically involves three steps. First, after demonstrating a significant effect of the independent variable X on the dependent variable Y (Step 1), the focus shifts to establishing a significant effect of the X on the mediation variable M (Step 2). Finally, the effect of the M on Y is examined. However, this method often fails to fully address endogeneity concerns that arise from including both the independent variable and the mediation variable in the same regression equation. Therefore, in this study, we adopt the approach from Jiang (2021) for testing mediation effects, emphasizing the identification of transmission

²In June 2017, the Chinese government designated the first group of Green Finance Reform and Innovation Pilot Zonocities, including Guiyang, Anshun, Quzhou, Huzhou, Nanchang, Jiujiang, Guangzhou, Changji, Hami and Karamay. In December 2019, Lanzhou was added to the pilot zone, followed by Chongqing in August 2022.

mechanisms while relying on strong theoretical and empirical foundations to support the significance of the third step (see Section 2.2.2). The regression results are presented in the table below.

Table 6 columns (1) present the mediation analysis results. Column (1) reveals a statistically significant positive association between *GFI* and corporate green development (*GD* is measured by green total factor productivity) at the 10% level. This demonstrates that *GFI* has a substantial positive effect on firms' green productivity. The model also controls for a comprehensive set of firm-specific characteristics and includes both industry and year fixed effects to account for unobserved heterogeneity. The mean value of *GD* is 1.003, therefore the coefficient of 0.052 can be interpreted as a one-unit change in *GFI* is associated with an approximate 5.18% ($0.052/1.003$) increase in green total factor productivity relative to its average level. This represents a substantial economic effect. The Adjusted R^2 of 0.0164 is acceptable for this type of analysis. The primary goal of this regression is not to fully explain the variance in *GTFP* but to obtain an unbiased estimate of the effect of *GFI* on the mediator. To sum up, Hypothesis 2 has been confirmed that the effect of green-driven risk mitigation does exist.

Table 6. Mechanism Analysis Results

Variables	(1) <i>GD</i>	(5) <i>GD</i>
<i>GFI</i>	0.052* (1.859)	0.009** (2.93)
<i>CR</i>		-0.002 (-1.20)
<i>GFI</i> × <i>CR</i>		0.00003*** (55.45)
<i>GTI</i>		
<i>GFI</i> × <i>GTI</i>		
<i>Controls</i>	YES	YES
IND FE	YES	YES
Year FE	YES	YES

Observations	3,664	3,664
Adjust R ²	0.0164	0.6505

5. Extended Analysis

5.1. Regional climate risk on corporate green development

Climate risks represent a primary challenge to biodiversity conservation (Bellard, 2012). Global warming triggered by climate change leads to glacial melting and sea-level rise, directly reducing the habitat areas of flora and fauna (Chen et al., 2012). Additionally, climate change disrupts species behaviors and threatens food chain stability (Urban, 2015). For example, plants are blooming earlier while insect life cycles fail to synchronize, disrupting pollination. Higher temperatures are also pushing some species toward the poles, where those unable to adapt face extinction, altering species composition and disturbing the ecological balance. Furthermore, climate change has emerged as a pivotal factor amplifying extreme climatic events' frequency and intensity (Newman & Noy, 2023). These events not only directly destroy wildlife habitats but also have widespread impacts on infrastructure and economic activities. Consequently, the detrimental effects of climate risks on biodiversity directly affect corporate operation decisions.

Climate risks primarily stem from greenhouse gas emissions caused by human activities (Ren et al., 2022). Specifically, green finance internalizes environmental costs into corporate production expenditures through carbon pricing mechanisms (e.g., emissions trading systems and carbon taxes), thereby incentivizing enterprises to enhance energy efficiency and reduce emissions (Zhang et al., 2022). Simultaneously, green finance provides financing tools, such as green bonds and credit, to finance corporate R&D initiatives in sustainable energy-efficient technology innovation, fostering advancements in technology and further supporting corporate energy conservation and emissions reduction (Lee & Lee, 2022). Furthermore, green finance drives the transition of energy systems to low-carbon models by funding the development of large-scale renewable energy infrastructure, particularly in solar and

wind power generation, effectively curbing carbon emissions and addressing climate-related risks.

Based on the above analysis, we hypothesize that extreme climate change in a firm's locality provides enterprises with direct exposure, thereby enhancing the transmission mechanism through which green finance promotes corporate green development. We thus introduce a regional climate risk index to conduct a moderating-effect analysis. Extending the methodological framework established by Guo et al. (2023), this paper develops a geospatial composite metric evaluating Regional Climate Risk (*CR*). The index synthesizes four meteorological hazard dimensions, which constitute Extreme Cold Days (ECD), Heatwave Days (HWD), Torrential Precipitation Events (TPE), and Severe Aridity Episodes (SAE).

The regression results from column (5) table 6 demonstrate a significant positive moderating role of regional climate risk (*CR*) in the relationship between green finance (*GFI*) and green development (*GD*). While the direct effect of climate risk itself is negative but statistically insignificant, the interaction term ($GFI \times CR$) is highly significant and positive. This indicates that the marginal effect of *GFI* on *GD* is strengthened in regions experiencing higher levels of climate risk. Consequently, rather than inhibiting progress, climate risk acts as a catalytic mechanism that enhances the efficacy of green finance initiatives. This finding suggests a powerful crisis-driven response mechanism. High climate risk generates a heightened demand for adaptation and mitigation, transforming green finance from a voluntary option into an urgent necessity. This underscores the critical importance of strategically directing green financial resources to high-risk regions, not merely as a compensatory measure for their vulnerability, but as a means to achieve economic returns on investment in the green transition.

5.2. Heterogeneity Analysis

5.2.1. Environmental Regulations

The effectiveness of green finance policy implementation may experience regional heterogeneity in legal and regulatory frameworks (Jin and Mengqi, 2011). Building upon the methodology of Yan et al. (2022), this study quantifies environmental regulatory stringency in the jurisdictions of sampled enterprises through a composite index incorporating. The sample firms were bifurcated into high-level and low-level environmental regulatory stringency groups based on the mean value of regulatory intensity, followed by subgroup regression analyses. As shown in Column (1) and Column (2) of Table 7, which present regression results for the high- and low-stringency groups respectively, the mitigating effect of the green finance policy on corporate biodiversity risks is significantly more pronounced in regions with stringent environmental regulations. This heterogeneous impact likely stems from synergistic effects between the green finance policy and preexisting regulatory frameworks, which collectively

5.2.2. Pollution Intensity

Enterprises with varying pollution levels face differentiated environmental compliance costs and green transition burdens. Following the National Economic Industry Classification standards, sampled firms were categorized into heavy-polluting (HP) and non-heavy-polluting (NHP) enterprise groups for subgroup regression. Table 7 reveals a statistically *BR* coefficient in Column (3) for NHP corporations, while it is not significant in Column (4) for HP corporations. The empirical results demonstrate that green finance instruments exert a pronounced suppressive impact on biodiversity risk exposure of NHP enterprises. One possible reason is that, compared to HP firms, NHP firms carry a lighter environmental burden and face less pressure from green transitions. The introduction of green finance can better guide them towards a more sustainable development path. For HP firms, while green finance can also bring positive changes, their high baseline pollution levels mean there is greater room for improvement, requiring more effort to achieve similar effects.

5.2.3. Executives' Green Awareness

Strategic cognition theory posits that managerial agency in corporate behavior reflects executives' perceptions and awareness of the external environment; managers make decisions driven by their cognitive interpretations of external stimuli, which in turn guide corporate actions (Florida & Davison, 2001). Variations in executives' green cognition may lead to differing corporate responses within the context of green finance. For instance, Maloney & Ward (1973) argue that the stronger an executive's ecological cognition—comprising environmental knowledge and awareness of environmental challenges—the more positive their environmental behaviors. Executives' environmental cognition in sampled enterprises is quantified using a composite metric incorporating approach from Chen et al. (2024). High and low green cognition groups are categorized by the mean value of executives' environmental cognition. Table 7 reports subgroup regression results by executive green awareness. Column (5) for the high-green-awareness subgroup shows a statistically significant negative coefficient for biodiversity risk (*BR*), while column (6) for the low-awareness subgroup displays an insignificant *BR* coefficient. The contrast confirms that green finance policies exert stronger mitigating effects on corporate biodiversity risks in firms with heightened environmental consciousness among executives. The underlying mechanism may be that in environments with well-developed green finance systems, firms led by executives with stronger green awareness tend to proactively seek green financial support to advance their environmental protection and sustainable development initiatives, thereby further mitigating biodiversity-related risks.

Table 7 Results of Heterogeneity Tests

	(1)	(2)	(3)	(4)	(5)	(6)
	High-level	Low-level	HP	NHP	High	Low
<i>GFI</i>	-2.673** (-2.509)	-0.104 (-0.176)	-1.529** (-2.446)	0.761 (1.007)	-1.130* (-1.837)	-0.654 (-0.859)
<i>Controls</i>	YES	YES	YES	YES	YES	YES
IND FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	1,314	2,378	2,803	854	2,332	1,716

Adjust R ²	0.218	0.143	0.142	0.220	0.170	0.175
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6. Conclusions and Discussion

This study employs panel data from Chinese listed firms (2000-2022) to assess green finance's influence on corporate biodiversity risks through econometric modeling. Empirical evidence confirms green finance significantly mitigates corporate biodiversity risk exposure, with statistical significance maintained, highlighting the interdependence between finance and biodiversity (Nedopil, 2023). Biodiversity-related risks disrupt production activities and value chains in the real economy, leading to asset impairment, weakened debt capacity, reputational harm and rising compliance costs, all of which threaten the stability and sustainable growth of the financial system (Kedward et al., 2023). Concurrently, the considerable funding gap in biodiversity conservation and risk management underscores the need for support from financial systems. Green finance, as a framework centered on ecological concerns, effectively channels social capital into environmentally favorable projects. In China, the government has emphasized biodiversity protection, embedding it within the green finance framework.

This study examined the mediating mechanisms by introducing corporate green development, measured by the green total factor productivity. Our firm-level regression analysis yields statistically significant support for Hypothesis 2. Green finance reduces corporate biodiversity exposure through enhanced green innovation capabilities, thereby corroborating the institutional innovation pathways identified in Hua et al. (2024). By adopting more energy-efficient production methods, green development reduces a firm's dependency on natural resources and minimizes environmental damage, thus directly lowering biodiversity-related physical risks (Zhao et al., 2024). Additionally, green development helps firms comply with regulatory standards and aligns with investor and consumer expectations, thereby reducing transition risks to biodiversity (Plumlee et al., 2015). At the regional level, empirical evidence indicates that regional climate risk serves as a moderating factor in the relationship between green

finance and corporate green development. It exerts a positive moderating effect that strengthens the facilitating impact of green finance on corporate green development. In other words, the influence of green finance becomes more pronounced when the level of climate risk is higher. It suggests that heightened exposure to climate-related threats can significantly amplify firms' intrinsic motivation to pursue green and low-carbon transformation. From a policy perspective, this result highlights the importance of designing green finance instruments that are responsive to regional climate conditions. Governments and financial institutions should not only expand the scale of green credit, bonds, and insurance products, but also prioritize the allocation of such resources to regions and sectors that face elevated climate risks. Furthermore, incorporating climate-risk metrics into financial decision-making frameworks can strengthen the resilience of enterprises while simultaneously advancing broader ecological and economic sustainability goals.

This study also conducted heterogeneity analysis basing on regional regulation intensity, industrial pollution intensity and executives' green awareness. The results reveals that the mitigating effect of green finance on corporate biodiversity risk is highly context-dependent. First, regional environmental regulation plays a crucial role: the impact of green finance is significantly stronger in jurisdictions with stringent regulatory frameworks, suggesting a synergistic effect between financial incentives and institutional enforcement. Second, at the industry level, green finance shows more pronounced effects among non-heavy-polluting enterprises. These firms, facing relatively lighter compliance costs and transition burdens, are better positioned to translate financial support into sustainable practices. By contrast, for heavy-polluting firms, the suppressive effect of green finance is weaker, reflecting the greater structural challenges and adjustment costs inherent in their transformation. Third, managerial cognition emerges as a decisive internal factor. Firms led by executives with heightened green awareness demonstrate a stronger capacity to leverage green finance for biodiversity risk management, highlighting the strategic role of leadership in shaping

corporate responses to external incentives. Collectively, these findings indicate that the effectiveness of green finance is conditioned by regulatory, industrial, and organizational heterogeneity. Tailoring green financial instruments to local regulatory contexts, industry characteristics, and managerial cognition is therefore essential to maximize their ecological and economic impact.

Notwithstanding its conceptual and practical contributions, this study remains circumscribed by some identifiable constraints that warrant methodological refinement in subsequent research. First, corporate biodiversity risk was assessed through sentiment analysis of listed firms' annual reports using Python, which may lead to biases if companies embellish their disclosures, potentially skewing the results. Future research could pursue more objective and scientifically robust measures of corporate biodiversity risk. Second, while the research employs data from Chinese listed firms, the country's geographical expanse creates substantial biogeographical variations, with marked ecological heterogeneity observable across distinct regional jurisdictions. Species richness and ecological vulnerability vary considerably, and this study does not account for these regional variations. Future studies could conduct a more granular analysis that incorporates regional distinctions. Third, a limitation of this study lies in the relatively small sample size. This issue mainly stems from the stringent criteria used to identify corporate biodiversity risk, as well as additional data loss during multi-source matching. As a result, we were unable to pursue further analyses, such as testing the synergistic effects of tax and financial policies. Future research may address this limitation by supplementing firm-level disclosures with unstructured, real-time data to broaden the empirical base.

Last, beyond the Green Finance Innovation Demonstration Zones, China has enacted multiple green policies, including low-carbon city pilots, ecological restoration initiatives and green supply chain management pilots, all of which may impact corporate biodiversity risk. This study does not include these policies, leaving room for further investigation into their effects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical statement

Not applicable. No use of human or animal samples or tissues.

Data availability

Data will be made available on request.

References

- Ardia, D., Bluteau, K., Boudt, K., & Inghelbrecht, K. (2023). Climate change concerns and the performance of green vs. brown stocks. *Management Science*, 69(12), 7607-7632. <https://doi.org/10.1287/mnsc.2022.4636>
- Bahl, S. (2012). Green banking-The new strategic imperative. *Asian Journal of Research in Business Economics and Management*, 2(2), 176-185. indianjournals.com/ijor.aspx?target=ijor:ajrbem&volume=2&issue=2&article=015
- Bellard, C., Bertelsmeier, C., Leadley, P., Thuiller, W., & Courchamp, F. (2012). Impacts of climate change on the future of biodiversity. *Ecology letters*, 15(4), 365-377. <https://doi.org/10.1111/j.1461-0248.2011.01736.x>
- Cardinale, B. J., Duffy, J. E., Gonzalez, A., Hooper, D. U., Perrings, C., Venail, P., ... & Naeem, S. (2012). Biodiversity loss and its impact on humanity. *Nature*, 486(7401), 59-67. <https://doi.org/10.1038/nature11148>
- Chen, C. C., McCarl, B., & Chang, C. C. (2012). Climate change, sea level rise and rice: global market implications. *Climatic change*, 110(3), 543-560. <https://doi.org/10.1007/s10584-011-0074-0>
- Chen, Y., Masron, T. A., & Mai, W. (2024). Role of investor attention and executive green awareness on environmental information disclosure of Chinese high-tech listed companies. *Journal of Environmental Management*, 365, 121552. <https://doi.org/10.1016/j.jenvman.2024.121552>
- Convention on Biological Diversity(CBD). (2022). Kunming-Montreal Global Biodiversity Framework (Draft Decision Submitted by the President). In Conference of the Parties to the Convention on Biological Diversity: Fifteenth Meeting-Part II (CBD/COP/15/L.25). Conference of the Parties (COP)
- Deutz, A., Heal, G. M., Niu, R., Swanson, E., Townshend, T., Zhu, L., ... & Tobin-de la Puente, J. (2020). Financing nature: Closing the global biodiversity financing gap. The Paulson Institute, The Nature Conservancy, and the Cornell Atkinson Center for

- Sustainability, 256. Financing Nature: Closing the Global Biodiversity Financing Gap - Paulson Institute
- Florida, R., & Davison, D. (2001). Gaining from green management: environmental management systems inside and outside the factory. *California management review*, 43(3), 64-84. <https://doi.org/10.2307/41166089>
- Fuchs-Schündeln, N., & Hassan, T. A. (2016). Natural experiments in macroeconomics. In *Handbook of macroeconomics* (Vol. 2, pp. 923-1012). Elsevier. <https://doi.org/10.1016/bs.hesmac.2016.03.008>
- Giglio, S., Kuchler, T., Stroebel, J., & Zeng, X. (2023). Biodiversity Risk. *NBER Working Paper*, 31137. [https:// DOI 10.3386/w31137](https://DOI.10.3386/w31137)
- Guo, K., Li, Y., Zhang, Y., Ji, Q., & Zhao, W. (2023). How are climate risk shocks connected to agricultural markets?. *Journal of Commodity Markets*, 32, 100367. <https://doi.org/10.1016/j.jcomm.2023.100367>
- Hu, J., Li, J., Li, X., Liu, Y., Wang, W., & Zheng, L. (2021). Will green finance contribute to a green recovery? Evidence from green financial pilot zone in China. *Frontiers in Public Health*, 9, 794195. <https://doi.org/10.3389/fpubh.2021.794195>
- Hua, M., Li, Z., Zhang, Y., & Wei, X. (2024). Does green finance promote green transformation of the real economy?. *Research in International Business and Finance*, 67, 102090. <https://doi.org/10.1016/j.ribaf.2023.102090>
- Huang, Y., Chen, C., Lei, L., & Zhang, Y. (2022). Impacts of green finance on green innovation: a spatial and nonlinear perspective. *Journal of Cleaner Production*, 365, 132548. <https://doi.org/10.1016/j.jclepro.2022.132548>
- Huang, Y., Xue, L., & Khan, Z. (2021). What abates carbon emissions in China: Examining the impact of renewable energy and green investment. *Sustainable Development*, 29(5), 823-834. <https://doi.org/10.1002/sd.2177>
- IPBES. (2019). Global assessment report on biodiversity and ecosystem services. Bonn: Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. <https://doi.org/10.5281/zenodo.3831673>
- Jiang, T. (2022). Mediating and Moderating Effects in Empirical Studies of Causal Inference. *China Industrial Economics*, 5, 100-120. <https://link.oversea.cnki.net/doi/10.19581/j.cnki.ciejournal.2022.05.005>
- Karolyi, G. A., & Tobin - de la Puente, J. (2023). Biodiversity finance: A call for research into financing nature. *Financial Management*, 52(2), 231-251. <https://doi.org/10.1111/fima.12417>
- Kedward, K., Ryan-Collins, J., & Chenet, H. (2023). Biodiversity loss and climate change interactions: financial stability implications for central banks and financial supervisors. *Climate Policy*, 23(6), 763-781. [10.1080/14693062.2022.2107475](https://doi.org/10.1080/14693062.2022.2107475)
- Komendantova, N. (2021). Transferring awareness into action: A meta-analysis of the behavioral drivers of energy transitions in Germany, Austria, Finland, Morocco, Jordan and Iran. *Energy Research & Social Science*, 71, 101826. <https://doi.org/10.1016/j.erss.2020.101826>
- Lee, C. C., & Lee, C. C. (2022). How does green finance affect green total factor

- productivity? Evidence from China. *Energy Economics*, 107, 105863. <https://doi.org/10.1016/j.eneco.2022.105863>
- Li, Y. H., & Huang, J. W. (2017). The moderating role of relational bonding in green supply chain practices and performance. *Journal of Purchasing and Supply Management*, 23(4), 290-299. <https://doi.org/10.1016/j.pursup.2017.06.001>
- Liang, C., Yang, J., Shen, L., & Dong, D. (2024). The role of biodiversity risk in stock price crashes. *Finance Research Letters*, 67, 105856. <https://doi.org/10.1016/j.frl.2024.105856>
- Lu, Y., Gao, Y., Zhang, Y., & Wang, J. (2022). Can the green finance policy force the green transformation of high-polluting enterprises? A quasi-natural experiment based on “Green Credit Guidelines”. *Energy Economics*, 114, 106265. <https://doi.org/10.1016/j.eneco.2022.106265>
- Lv, C., Bian, B., Lee, C. C., & He, Z. (2021). Regional gap and the trend of green finance development in China. *Energy Economics*, 102, 105476. <https://doi.org/10.1016/j.eneco.2021.105476>
- Ma, F., Wu, H., & Zeng, Q. (2024). Biodiversity and stock returns. *International Review of Financial Analysis*, 103386. <https://doi.org/10.1016/j.irfa.2024.103386>
- Maloney, M. P., & Ward, M. P. (1973). Let's Hear from the people, an objective scale for the measurement of ecological attitudes and knowledge. *American Psychologist*, 30(787.790). <https://doi.org/10.1037/h0034936>
- Ministry of Environment Protection of the People's Republic of China (MEPC). (2023). China Biodiversity Conservation Strategy and Action Plan (2023-2030). MEE. China-Biodiversity-Action-Plan-1.pdf
- Nedopil, C. (2023). Integrating biodiversity into financial decision-making: Challenges and four principles. *Business Strategy and the Environment*, 32(4), 1619-1633. <https://doi.org/10.1002/bse.3208>
- Newman, R., & Noy, I. (2023). The global costs of extreme weather that are attributable to climate change. *Nature Communications*, 14(6103). 10.1038/s41467-023-41888-1
- NGFS & INSPIRE. (2021). Biodiversity and financial stability: exploring the case for action, NGFS Occasional paper. Biodiversity and financial stability: exploring the case for action
- Pörtner, H. O., Scholes, R. J., Agard, J., Archer, E., Arneth, A., Bai, X., ... & Ngo, H. T. (2021). IPBES-IPCC co-sponsored workshop report on biodiversity and climate change. Bonn: IPBES and IPCC. <https://doi.org/10.5281/zenodo.4782538>
- Plumlee, M., Brown, D., Hayes, R. M., & Marshall, R. S. (2015). Voluntary environmental disclosure quality and firm value: Further evidence. *Journal of Accounting and Public Policy*, 34(4), 336-361. <https://doi.org/10.1016/j.jaccpubpol.2015.04.004>
- Ren, X., Li, Y., Shahbaz, M., Dong, K., & Lu, Z. (2022). Climate risk and corporate environmental performance: Empirical evidence from China. *Sustainable Production and Consumption*, 30, 467-477. <https://doi.org/10.1016/j.spc.2021.12.023>
- Rockström, J., Steffen, W., Noone, K., Persson, Å., Chapin, F. S., Lambin, E. F., ... &

- Foley, J. A. (2009). A safe operating space for humanity. *Nature*, 461(7263), 472–475. <https://doi.org/10.1038/461472a>
- Scholtens, B. (2017). Why finance should care about ecology. *Trends in Ecology & Evolution*, 32(7), 500-505. 10.1016/j.tree.2015.06.015
- Seidl, A., Cumming, T., Arlaud, M., Crossett, C., & van den Heuvel, O. (2024). Investing in the wealth of nature through biodiversity and ecosystem service finance solutions. *Ecosystem Services*, 66, 101601. <https://doi.org/10.1016/j.ecoser.2024.101601>
- Smith, T., Beagley, L., Bull, J., Milner-Gulland, E. J., Smith, M., Vorhies, F., & Addison, P. F. (2020). Biodiversity means business: Reframing global biodiversity goals for the private sector. *Conservation Letters*, 13(1), e12690. <https://doi.org/10.1111/conl.12690>
- Steffen, W., Richardson, K., Rockström, J., Cornell, S. E., Fetzer, I., Bennett, E. M., ... & Sörlin, S. (2015). Planetary boundaries: Guiding human development on a changing planet. *Science*, 347(6223), 1259855. <https://doi.org/10.1126/science.1259855>
- Urban, M. C. (2015). Accelerating extinction risk from climate change. *Science*, 348(6234), 571-573. 10.1126/science.aaa4984
- Wang, L., Yang, X., & Cai, Q. (2024). Influence mechanism of green finance on regional emission reduction. *Heliyon*, 10(1). 10.1016/j.heliyon.2023.e23861
- Wang, Z., Wang, X., Wang, Z., & Mai, S. (2024). The impact of green low-carbon development on public health: a quasi-natural experimental study of low-carbon pilot cities in China. *Frontiers in Public Health*, 12, 1470592. <https://doi.org/10.3389/fpubh.2024.1470592>
- WEF (World Economic Forum). (2020). New Nature Economy Report II, The Future of Nature and Business. World Economic Forum, Cologny/Geneva, Switzerland. New Nature Economy Report II: The Future Of Nature And Business | World Economic Forum
- White, T. B., Petrovan, S. O., Bennun, L. A., Butterworth, T., Christie, A. P., Downey, H., ... & Sutherland, W. J. (2023). Principles for using evidence to improve biodiversity impact mitigation by business. *Business Strategy and the Environment*, 32(7), 4719-4733. <https://doi.org/10.1002/bse.3389>
- Wilting, H. C., & van Oorschot, M. M. (2017). Quantifying biodiversity footprints of Dutch economic sectors: A global supply-chain analysis. *Journal of Cleaner Production*, 156, 194-202. <https://doi.org/10.1016/j.jclepro.2017.04.066>
- World Economic Forum. (2020). New nature economy report II: The future of nature and business. Geneva: WEF. Retrieved from <https://www.weforum.org/reports/new-nature-economy-report-ii-the-future-of-nature-and-business>
- Wu, G., Liu, X., & Cai, Y. (2024). The impact of green finance on carbon emission efficiency. *Heliyon*, 10(1). <https://doi.org/10.1016/j.heliyon.2023.e23803>
- Wu, H., Li, Y., Hao, Y., Ren, S., & Zhang, P. (2020). Environmental decentralization, local government competition, and regional green development: Evidence from

- China. *Science of the Total Environment*, 708, 135085. <https://doi.org/10.1016/j.scitotenv.2019.135085>
- Xing, C., Zhang, Y., & Tripe, D. (2021). Green credit policy and corporate access to bank loans in China: The role of environmental disclosure and green innovation. *International Review of Financial Analysis*, 77, 101838. <https://doi.org/10.1016/j.irfa.2021.101838>
- Xu, J., Wang, J., Li, R., & Gu, M. (2023). Is green finance fostering high-quality energy development in China? A spatial spillover perspective. *Energy Strategy Reviews*, 50, 101201. <https://doi.org/10.1016/j.esr.2023.101201>
- Yan, C., Di, D., Li, G., & Wang, J. (2022). Environmental regulation and the supply efficiency of environmental public services: Evidence from environmental decentralization of 289 cities in China. *Growth and Change*, 53(2), 515-535. <https://doi.org/10.1111/grow.12596>
- Zhao, S., Zhang, L., Peng, L., Zhou, H., & Hu, F. (2024). Enterprise pollution reduction through digital transformation? Evidence from Chinese manufacturing enterprises. *Technology in Society*, 77, 102520. <https://doi.org/10.1016/j.techsoc.2024.102520>
- Zhang, D., Mohsin, M., & Taghizadeh-Hesary, F. (2022). Does green finance counteract the climate change mitigation: Asymmetric effect of renewable energy investment and R&D. *Energy Economics*, 113, 106183. <https://doi.org/10.1016/j.eneco.2022.106183>
- Zhang, Z., Shi, K., Gao, Y., & Feng, Y. (2023). How does environmental regulation promote green technology innovation in enterprises? A policy simulation approach with an evolutionary game. *Journal of Environmental Planning and Management*, 68(5), 979–1008. <https://doi.org/10.1080/09640568.2023.2276064>
- Zhang, Z., Hua, C., Song, M., Jiang, M. S., Miao, J., & Chen, X. L. (2025). Evaluation and influencing factors of sustainable economic development: a spatial analysis in the yellow river basin. *Economic Analysis and Policy*, 85(3), 997-1011. <https://doi.org/10.1016/j.eap.2025.01.012>
- Zhang, Z., Wang, J., Feng, C., & Chen, X. (2023). Do pilot zones for green finance reform and innovation promote energy savings? evidence from china. *Energy Economics*, 124, 106763. <http://dx.doi.org/10.1016/j.eneco.2023.106763>
- Zhang, Z., Li, P., Wang, X., Ran, R., & Wu, W. (2024). New energy policy and new quality productive forces: a quasi-natural experiment based on demonstration cities. *Economic Analysis & Policy*, 84, 1670. <https://doi.org/10.1016/j.eap.2024.10.039>

Highlights

- Based on text mining and BERT model, a sentiment-based index to quantify firm-level biodiversity risk is constructed.
- Green finance significantly curbs biodiversity risk via green development and climate risk reduction.
- Green finance is more effective in low-pollution firms and with high executive green awareness.
- Conclusions provide empirical support for using green finance as a strategic tool for biodiversity risk management.